Novel aadA5 and dfrA17 variants of class 1 integron in multidrug-resistant Escherichia coli causing bovine mastitis

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ENVIRONMENTAL BIOTECHNOLOGY

Novel *aadA5* and *dfrA17* variants of class 1 integron in multidrug-resistant *Escherichia coli* causing bovine mastitis

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Abstract

Mobile genetic elements (MGEs) are associated with the emergence of multidrug resistance in extended-spectrum β-lactamase (ESBL)-producing *Enterobacteriaceae*. This study explores the role of class 1 integrons and IS*26* elements in breaching taxonomic barriers. A total of 110 *E. coli* bacteria were isolated from 300 clinical mastitis milk samples. The 98% *E. coli* isolates were extended-spectrum beta-lactamase- producers. About 83% of these isolates carried co-resistance for fluoroquinolones. The co-existence of (extended-spectrum beta-lactamase + quinolone resistance determining region mutations) and (extended-spectrum beta-lactamase + plasmid-mediated quinolone resistance genes) was found in 76% and 44% of isolates, respectively. The MGEs were detected in 88% of isolates with IS*26* in 82% and class 1 integrase in 40% of isolates. The types of class 1 integron gene cassettes detected includes *dfrA7*, (*dfrA17* + *aadA5*), and (*dfrA1* + *aadA1*). We discovered 2 and 4 novel variants of the *dfrA17* and *aadA5* genes, respectively. We report a variant of *aadA5* with mutation E235G in the Indian subcontinent earlier reported only in a human clinical isolate from Belgium. About 19 isolates carried IS*26* linked to integrase gene *intI1* with an internal deletion of 265 bp in the 5`CS of integrase gene *intI1*, earlier reported only in *E. coli* ST131 isolates from human clinical, wastewater samples. This study suggests intercontinental dissemination of antibiotic resistant genes (ARGs) across different microbiomes via mobile genetic elements.

Key points

- *The role of mobile genetic elements in the emergence of multidrug-resistant E. coli in bovine mastitis.*
- *Novel variants of the aadA5 (aminoglycoside adenyl transferase) and dfrA17 (dihydrofolate reductase) genes were identified in pathogenic E. coli isolated from bovine mastitis in class 1 integron gene cassette.*
- *Sequence analysis of mobile genetic components revealed the physical connection between IS26 and intI1 genes with an internal deletion in 5'CS of class 1 integrase.*

Keywords Class 1 integrons · IS element · Inter-kingdom ARGs transfer · Mobile genetic elements · Multidrug resistance

Introduction

Antimicrobial resistance (AMR) is a global concern associated with humans, animals, and the environment (Lomazzi et al. 2019). AMR is the occurrence of resistance to antimicrobials like antibiotics, antifungals, or antivirals in infectious microorganisms (bacteria, fungi, parasites) (Blaskovich 2018). In 2019, World Health Organization (WHO) declared AMR as top 10 global health threats as AMR makes treatment of infections difficult and costlier ("Global Antimicrobial Resistance and Use Surveillance System (GLASS) Report," 2021). The increasing rate of AMR can be attributed to misuse/overuse of antimicrobials by humans and agricultural utilization in the feed of food animals for growth enhancement, disease prevention, and increasing feed capacity (Samreen et al. 2021). Other than the injudicious use of antimicrobials transfer of antibiotic resistant genes (ARGs) through horizontal gene transfer (HGT) is a major concern. The phenomenon of HGT

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resulted in the accumulation of multiple ARGs belonging to different antibiotic classes in a single bacteria causing the emergence of multidrug resistance (MDR) (Davies en Davies 2010). Bacteria carrying different plasmids with multiple ARGs are disposed into the environment from clinical and livestock waste. This discharge of antibiotic resistant bacteria (ARB) makes the environment an important ally between humans and animals for the exchange of genetic material (Manaia 2017). The occurrence of MDR bacteria in human, animal, food industries, and the natural environment is threatening and costly in terms of morbidity and mortality (Sun et al. 2019). The transmission of ARB from the environment and high antibiotic consumption in livestock culminate in the emergence of MDR in bacteria causing bovine mastitis (Liang et al. 2021). Bovine mastitis, one of the costliest diseases in the dairy industry leads to economic losses as the quality and quality of the milk gets compromised adding degradation to the health of affected cows (Halasa et al. 2011). It is the inflammation in mammary glands due to the invasion of pathogenic bacteria in the udder of cattle (Bradley 2002). *Escherichia coli* is the major gram-negative bacterium causing clinical and subclinical mastitis in dairy cows acquired through the environment (Ruegg 2021). The most effective treatment considered for *E. coli* infections is antibiotic therapy (Gomes en Henriques 2015). The most common antibiotics used are third and fourth generation cephalosporins (Suojala et al. 2013). These extended-spectrum cephalosporins were soon affected by the emergence of extended-spectrum β-lactamases (ESBLs) and AmpC β-lactamases (Eriksen en Blanco 1992). The major ESBLs include bla_{TEM}, bla_{SHV}, bla_{OXA}, and $bla_{\mathrm{CTX\text{-}M}}$ (Bradford 2001). bla_{TEM} and $bla_{\mathrm{CTX\text{-}M\text{-}1}}$ group are considered the most prevalent ESBL globally and in major parts of Asia (Ogutu et al. 2015). The ESBL-producing *E. coli* are now emerging as MDR, carrying co-resistance to antibiotics like fluoroquinolones, trimethoprim/sulfamethoxazole, and aminoglycosides (Filioussis et al. 2020). The AWaRe classification of antibiotics by WHO considers cephalosporins and fluoroquinolones as watch antibiotics with a relatively high risk of bacterial resistance ("2021 AWaRe classification," WHO). Fluoroquinolones (FQs) are widely used as they display a broad antibacterial spectrum (Smith et al. 2012). The overuse of FQs led to the emergence of fluoroquinolone resistance owing to both vertical and horizontal transmission mechanisms (Kotb et al. 2019). Fluoroquinolone resistance is commonly mediated by mutations in the quinolone resistance-determining region (QRDR) of target genes *gyrA*, *gyrB* (DNA gyrase), and *parC*, *parE* (topoisomerase IV) (Hooper 2001). The vertical transmission of FQs resistance takes place through plasmid-mediated quinolone resistance (PMQR) genes including qnr genes (*qnrA*, *qnrB*, *qnrC*, *qnrD*, and *qnrS*), *aac(6')-Ib-cr*, and plasmid-mediated efflux pumps (*oqxAB*, *qepA*) (Jacoby et al. 2015). The co-existence of ESBL with PMQR and QRDRs has been reported globally (Shibl et al. 2012) and is of major concern as the presence of mobile genetic elements (MGEs) in these isolates leads to the establishment of MDR. MGEs like insertion sequences (IS), plasmids, transposons (Tn), and integrons (intI) are responsible for harboring ARGs and transmitting them across microbiomes (Partridge et al. 2018). Class 1, 2, and 3 integrons are the three main types of integrons associated with antimicrobial resistance (Cambray et al. 2010). Class 1 integrons possess conserved regions known as 5` conserved segment (5`CS) and 3` conserved segment (3`CS). The 5`CS includes gene for class 1 integrase (*intI1*), recombination site (attI), and a promoter (Pc). Whereas the 3`CS carries ARGs genes qacEΔ1 and *sul1* imparting resistance to quaternary ammonium compounds and sulfonamides, respectively (Fluit en Schmitz 1999). The diversification among integrons should be attributed to the presence of a variety of gene cassettes in them. Around 4000 to 18,000 unique gene cassettes have been detected per 0.3 g of soil indicating the richness of gene cassettes (Ghaly et al. 2019). The association of ARGs with the gene cassettes conferring resistance to most classes of antibiotics like aminoglycosides, trimethoprim, chloramphenicol, and beta-lactams displays its affluence in HGT (Gatica et al. 2016). The co-existence of ARGs with class 1 integrons, gene cassettes and IS elements like IS*26* may expedite the transmission of multiple ARGs into different microbiomes. MGEs and ARGs association with environmental resistome makes bacteria adaptable to survive in harsh conditions and increases their spread (Wellington et al. 2013). This is high time to obtain an understanding of the diverse transmission pathways of ARGs associated with MGEs which are likely to be in a loop between the human and animal communities. This study focuses on the co-resistance of fluoroquinolones in ESBL-producing *E. coli* in bovine mastitis. The presence of mobile genetic elements and their spread across human and animal microbiome was also noticed.

Materials and methods

Sample collection

A total of 300 milk samples were collected from lactating cows diagnosed with clinical mastitis in Karnal, Haryana (A northern state in India), and its nearby regions between January 2019 to April 2021. The samples were selected from cattle with decreased milk production, color change of the milk, and inflammation of udders. Clinical mastitis is determined using the California mastitis test (Ruegg en Reinemann 2002). Infected milk samples were collected aseptically to avoid any possible contamination while transport and storage.

Isolation of bacteria

The clinical mastitis milk samples stored at 4 °C after collection were directly streaked on MacConkey agar (MH081, HiMedia, India) and incubated overnight at 37 °C for the growth of lactose fermenting *Enterobacteriaceae*. Individual colonies with a characteristic pink-red color were selected representing *E. coli* bacterium. Isolated colonies selected from MacConkey agar were further streaked on HiCrome™ ESBL Agar (M1829, HiMedia, India) supplemented with 4 µg/ml cefotaxime (CTX) and 4 µg/ml ciprofloxacin (CIP). A total of 110 cefotaxime and ciprofloxacin-resistant isolates with characteristic pink to purple colors were selected. *E. coli* isolates were preserved at − 80 °C in Luria–Bertani (LB) broth supplemented with 80% glycerol.

Antimicrobial susceptibility testing

Antimicrobial susceptibility testing (AST) was performed on Mueller–Hinton agar (MHA) using the standard disc diffusion (DD) method. The overnight cultures of all 110 *E. coli* isolates were prepared in LB broth, adjusted to 0.5 McFarland standard, and tested with antibiotic discs (HiMedia, India) cefotaxime (30 µg), ciprofloxacin (5 µg), gentamicin (10 µg), streptomycin (10 µg), co-trimoxazole (25 µg), imipenem (10 µg) (Fig. S1). The isolates were grouped as resistant, susceptible, or intermediate as per Clinical & Laboratory Standards Institute guidelines.

DNA extraction and phylogenetic grouping of E. coli isolates

Genomic DNA was extracted using phenol–chloroform and purified by ethanol precipitation method as described earlier (De et al. 2010). The *E. coli* isolates were further confirmed for the presence of *E. coli-specific uidA* gene using PCR. The *E. coli* isolates were distributed into phylogenetic groups A, B1, B2, C, D, E, and F by PCR, as previously described (Clermont et al., 2013). The primers used are described in Table S1.

Detection of antimicrobial resistance genes

The *E. coli* isolates resistant to CTX and CIP were tested for the presence of β-lactamase (*bla*) encoding genes ($bla_{CTX\text{-}M\text{-}1}$ group, $bla_{CTX\text{-}M\text{-}9}$ group, $bla_{CTX\text{-}M\text{-}2}$ group, $bla_{CTX\text{-}M\text{-}25}$ group, bla_{SHV}, bla_{TEM}, $bla_{OXA\text{-}48}$, $bla_{OXA\text{-}1}$, and *AmpC* CMY-2) and PMQR genes (*qnrA*, *qnrB*, *qnrD*, *qnrS*, *aac(6′)-Ib-cr*, *qepA*, *oqxA*, and *oqxB*) by PCR using specific primers described in Table S1. The PCR conditions were as follows: pre-denaturation at 95 °C for 5 min; 30 cycles of denaturation at 95 °C for 30 s, annealing at 60 °C for 30 s, extension at 72 °C for 30 s; 72 °C extension for 5 min. The amplified products were separated on 1.5–2% agarose gel by electrophoresis, and ethidium bromide-stained gel was analyzed under Gel Doc—Gel Documentation System.

Detection of QRDR mutations

The frequently reported mutations in the QRDR regions of genes *gyrA* (S83L/D87N), *gyrB* (S492N), *parC* (S80I), and *parE* (S458A) were detected using allele-specific primers. The primers were designed to analyze allelic variants with 3`base of forward primer matching only with mutated allelic base of *gyrA* 248 (C → T), *gyrA* 259 (G → A), *gyrB* 1475 (G → A), *parC* 239 (G → T), *parE* 1372 (T → G). A reverse primer for each was designed downstream of the polymorphic site with no SNPs. The amplified products were separated on 2% agarose gel by electrophoresis, and ethidium bromide-stained gel was analyzed under Gel Doc—Gel Documentation System.

Determination of MICs

The minimum inhibitory concentration of cefotaxime and ciprofloxacin was determined using HiComb™ MIC Strip (MD064 and MD017) (HiMedia, India). Five individual colonies of *E. coli* harboring different combinations of ESBL and fluoroquinolone resistant genes were inoculated in LB broth, adjusted to 0.5 McFarland standard, and spread on MHA plates. The HiComb strips were applied to the agar surface and incubated for 18–24 h at 37 °C. The minimum inhibitory concentration (MIC) value was defined as the lowest concentration of the antibiotics giving rise to no visible growth.

Detection of mobile genetic elements

A total of 92 *E. coli* isolates were found to carry molecular factors imparting resistance against both β-lactam and fluoroquinolones. Those isolates were screened for the presence of mobile genetic elements that may be involved in the horizontal transfer of these resistant factors. Twelve different mobile genetic elements were checked by PCR using primers described in Table S1.

Detection and characterization of antibiotic resistance gene cassettes in class 1 integrons

The class 1 integron positive isolates were screened through PCR for the presence of variable regions (gene cassettes) with 5`CS and 3`CS primers described in Table S1. The gene cassette amplicons were subjected to Restriction fragment length polymorphism (RFLP) with restriction enzyme *HinfI* to determine if all the amplicons have the same gene composition. For RFLP, a 50 µl mixture containing 45 µl of the PCR product, 1 µl of *HinfI* (Thermo Scientific, ER0801), and 5 µl of 10 X digestion buffer supplied by the manufacturer was incubated at 37° C for 30 min. The digested product was run in 3% agarose gel stained with ethidium bromide to visualize bands in Gel Doc—Gel Documentation System. The PCR products with the same RFLP pattern were considered to contain the same gene cassette. To analyze the sequences of these gene cassettes, an amplicon representative of each restriction profile was subjected to sanger sequencing. The nucleotide sequences obtained were analyzed and *compared* to sequences submitted in databases by *Bla*stN. Furthermore, the presence of sulfonamide resistant gene (*sul1*) and *qacΔE1* conferring resistance against quaternary ammonium compounds associated with 3`CS of class 1 integron were also detected using PCR.

Sequencing and analysis of dfrA17 + aadA5 gene cassette

To further analyze the gene cassette with composition *dfrA17* + *aadA5*, 12 isolates were randomly selected. The integron cassette was amplified with primers 5`CS and 3`CS and cloned into pJET1.2/blunt Cloning Vector (CloneJET PCR Cloning Kit, K1232). Recombinant plasmid DNA was subjected to sanger sequencing. The nucleotide sequences obtained were analyzed and compared to the sequences submitted in the Bacterial Antimicrobial Resistance Reference Gene Database for *dfrA17* (www.ncbi.nlm.nih.gov/pathogens/refgene/#dfrA17) and *aadA5* (www.ncbi.nlm.nih.gov/pathogens/refgene/#aadA5) by *Bla*stN and *Bla*stP.

MIC determination of new variants of dfrA17

To compare the activity of native dihydrofolate reductase (*dfrA17*) and its new variants, we determined the minimum inhibitory concentration of trimethoprim using HiComb™ MIC Strip (MD021). Three individual colonies of *E. coli* harboring native and new variants of *dfrA17* resistant gene were inoculated in LB broth, adjusted to 0.5 McFarland standard, and subjected to MIC test.

Analysis of deleterious and tolerant substitutions by SIFT and PROVEAN

The potential effect of the variations in *aadA5* and *dfrA17* genes were predicted using SIFT software (https://sift.bii.a-star.edu.sg/www/SIFT_seq_submit2.html) and Provean software (http://provean.jcvi.org/seq_submit.php). SIFT (Sorting Intolerant from Tolerant) determines whether an amino acid substitution affects protein function based on sequence homology and the physical properties of amino acids. An amino acid substitution with a score less than 0.05 in SIFT software is considered deleterious.

PROVEAN (Protein Variation Effect Analyzer) analyzes if substitution is going to affect the biological function of protein using evolutionary conservation in different species. An amino acid substitution is considered to show a deleterious effect if the PROVEAN score is equal to or below a predefined threshold (e.g., − 2.5).

Molecular docking of aminoglycoside adenyl transferase (aadA5)

To study the effect of variants of *aadA5* on streptomycin, we conducted molecular docking studies. The 3D structure of aminoglycoside adenyl transferase (*aadA5*) protein was modeled using the SWISSMODEL server. Template 5g5a.1.A was selected for modeling protein with 43% sequence identity and 100% coverage. Verification of the selected model for structural constraints was conducted using a Ramachandran plot generated with the PROCHECK Server (https://servicesn.mbi.ucla.edu/PROCHECK/). The modeled structure showed 93.5% residues in the most favored regions of the Ramachandran plot. The K43R, (I31L + L33A), (D45V + I31M), F122S, and E235G variants of *aadA5* were constructed by incorporation of mutations in native protein using PyMOL (https://pymol.org/2/). The structure of ligand streptomycin with identification ID 19,649 was downloaded from NCBI PubChem. Docking was performed with the ligand using Autodock4.2 for native and new variants. The binding affinity between the ligand and protein were calculated using the AutoDock Vina (Trott en Olson 2010). Docking results were analysed using PyMOL and Maestro (Schrödinger Release 2021–4: Maestro, Schrödinger, LLC, New York, NY, 2021).

Results

Identification of β-lactams and fluoroquinolones resistant E. coli

A total of 110 *E. coli* isolates were obtained from 300 clinical mastitis milk samples and phenotypically characterized for antibiotic resistance against β-lactams and fluoroquinolones on HiCrome™ ESBL Agar (M1829, HiMedia, India). Antimicrobial susceptibility testing revealed that the majority of isolates were resistant to trimethoprim/sulfamethoxazole (90%), streptomycin (70%), and imipenem (90%) along with cefotaxime (100%), and ciprofloxacin (100%) (Fig. S2). Comparatively, 80% of the isolates were observed in the intermediate category against gentamicin. The presence of the *uidA* (β-D-glucuronidase) gene in the genomic DNA of all 110 bacterial isolates was detected by PCR and thus confirmed these organisms as *E. coli*. The phylogenetic group of each isolate was assigned as described earlier (Clermont et al. 2013). Among 110 isolates majority belonged to phylogroup A (32%) and B1 (31%) while the others belonged to D (8%), C (2%), F (2%), and B2 (1%). The 2% of the isolates belonged to group clade I or II whereas 22% of isolates could not be assigned to any phylogroup.

Prevalence of ESBL and PMQR genes

The occurrence of ESBL-producing genes was detected in 98% (108/110) *E. coli* isolates. The high prevalence of $bla_{\text{CTX-M}}$ in 95% (105/110) was observed which comprised 71% (79/110) $bla_{\text{CTX-M-1}}$ and 23% (26/110) $bla_{\text{CTX-M-9}}$. Other ESBL-producing genes identified were bla_{TEM}, *AmpC* CMY-2, and $bla_{\text{OXA-1}}$ in 86% (95/110), 30% (33/110), and 12% (14/110) of the isolates respectively (Fig. 1a). None of the isolates were found to carry ESBL genes $bla_{\text{CTX-M-2}}$ group, $bla_{\text{CTX-M-25}}$ group, bla_{SHV}, or $bla_{\text{OXA-48}}$. Among the 108 ESBL-producing isolates, 45% (49/108) were found to co-express PMQR genes providing resistance against fluoroquinolones (Fig. 2). The prevalence of *qnrS* was highest among plasmid-mediated quinolone-resistant determinants. The 79% (39/49) isolates were positive for *qnrS* followed by the presence of *aac(6′)-Ib-cr* in 24% (12/49) and *qnrB* in 12% (6/49) isolates. The *qnrA*, *qnrD*, *qepA*, *oqxA*, and *oqxB* were not detected in any of the isolates.

Genotypic detection of the QRDR determinants

The chromosomal QRDR mutations were screened in 110 CTX and CIP resistant *E. coli* isolates. The 76% (84/110) of isolates were found to carry at least one QRDR mutation. The most prevalent point mutations observed was *gyrB* (S492N) in 61% isolates (Fig. 1b) followed by *parC* (S80I, 46%), *gyrA* (S83L, 41%), *gyrA* (D87N, 40%), and *parE* (S458A, 4%) (Fig. S3–S7). There were 82% (69/84) isolates containing more than one QRDR mutation. The presence of two QRDR mutations was found in 32% (27/84) isolates with the majority carrying *gyrA* (D87N) + *gyrB* (S492N) in combination. There were 24/84 (28%), 15/84 (17%) and 3/84 (3%) isolates carrying three, four, and five QRDR mutations in combination (Fig. S8). The co-existence of QRDR and PMQR was observed in 37% (41/110) isolates. The MIC results revealed that the isolates harbored all five QRDR mutations (*gyrA*(S83L/D87N), *gyrB*(S492N), *parC*(S80I), *parE*(S458A) alone or in combination with the PMQR gene) displayed higher MIC than isolates carrying only PMQR gene. We also observed that the isolates with QRDR mutations (*gyrA*(S83L/D87N) + *gyrB*(S492N) + *parC*(S80I) + *parE*(S458A) alone or in combination with aac(6′)-Ib-cr) displayed higher MIC of 120 µg/ml, while the presence of PMQR alone lowered the MIC to 60 µg/ml (Fig. S9).

MGEs in E. coli isolates

In this study, six different mobile genetic elements were detected among 88% (81/92) isolates: IS*26* (82%, 76/92), *intI1* (40%, 36/92), IncFIA (30%, 28/92), IncFIB (33%, 31/92), IncFII (41%, 38/92) and IncI1 (7%, 7/92). Among the 36 isolates carrying *intI1* in 52% (19/36), *intI1* was found physically linked to the IS*26*. The PCR with IS*26* and *intI1* forward primers (Table S1) resulted in PCR amplicon of size ~ 1200 bp, suggesting that both the genes are in the same direction. The PCR product of three randomly selected isolates representing amplicon of 1200 bp was sent for Sangar sequencing. The sequencing results are submitted to NCBI GenBank under accession numbers OM735567, OM735568, and OM735569. A total of 20 different combinations (Table 1) of mobile genetic elements was observed with IS*26* being predominant (37%, $n = 30$) followed by IS*26* + *intI1* + IncFIA + IncFIB + IncFII (17%,

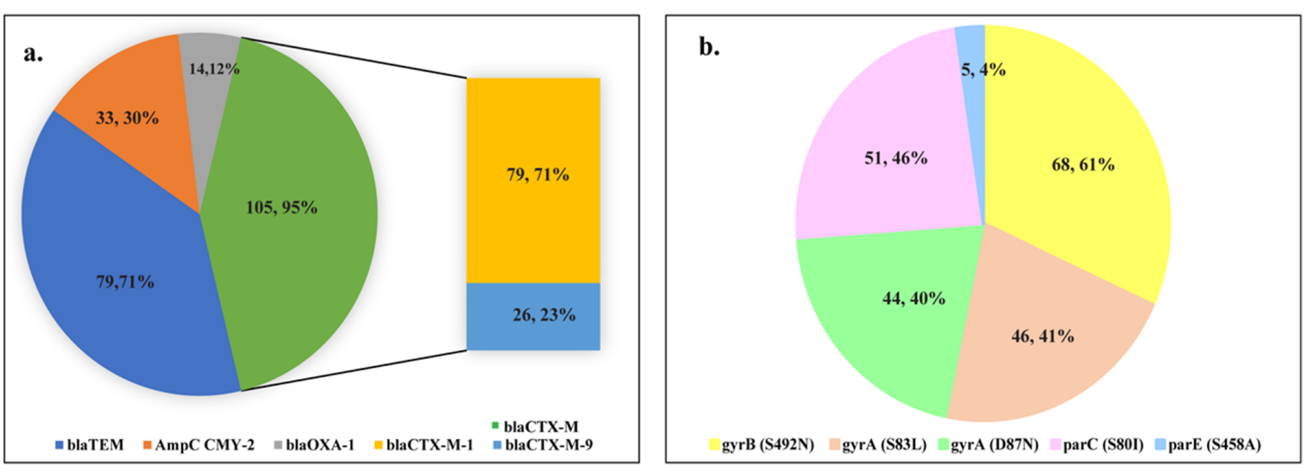

Fig. 1 Graphical representation of the % distribution frequencies. **a** ESBL producing genes among 110 *E. coli* isolates. **b** QRDR mutations among 110 *E. coli* isolates

Fig. 2 The distribution of antimicrobial resistance genes and mobile genetic elements among 92 β-lactam and fluoroquinolone resistant *E. coli* isolates belonging to different phylogroups represented by boxes of different color. Red, phylogroup B1; green, phylogroup A; blue, phylogroup C; black, phylogroup D; orange, phylogroup F; yellow, phylogroup B2; violet, clade I/II; pink, unidentified

$n = 14$). All isolates were negative for *intI2*, *intI3*, IncFIC, IncB/O, IncHI2, and IncK/B. While in 11 isolates none of the mobile genetic elements were observed.

Table 1 Composition of mobile genetic elements in 92 β-lactam and fluoroquinolone-resistant *E. coli* isolates from bovine mastitis

S.No	Composition of mobile genetic elements	Number (percentage)
1	IS*26*	30 (37)
2	IS*26* + *intI1* + IncFIA + IncFIB + IncFII	14 (17)
3	IS26 + *intI1*	8 (9)
4	IS26 + IncFIB + IncFII	5 (6)
5	IS26 + *intI1* + IncFIA + IncFII	4 (5)
6	IS26 + IncFIB + IncFII + IncI1	2 (2)
7	IS26 + IncFIA + IncFIB + IncFII	2 (2)
8	IS26 + IncFII	2 (2)
9	IS26 + *intI1* + IncFIA + IncFII + IncI1	2 (2)
10	IS26 + *intI1* + IncFII	2 (2)
11	IS26 + *intI1* + IncFII + IncI1	1 (1)
12	IS26 + *intI1* + IncFIB	1 (1)
13	IS26 + *intI1* + IncFIA + IncFIB	1 (1)
14	IS26 + IncFIA	1 (1)
15	IncFIB + IncI1	1 (1)
16	IS26 + IncFIB	1 (1)
17	IncFIB + IncFII + IncI1	1 (1)
18	IncFIA	1 (1)
19	IncFIB	1 (1)
20	*intI1*	1 (1)

Class 1 integron gene cassettes

The class 1 integron gene cassettes carry an array of antibiotic resistance genes, which we detected using 5`CS and 3`CS flanking the variable region. Among 36 *intI1* gene positive *E. coli*, 91% (33/36) of the isolates carried gene cassettes. Gene cassettes of two variable sizes ~ 1600 bp and ~ 800 bp (Fig. S10) were obtained and subjected to PCR–RFLP. Three different PCR–RFLP profiles of gene cassettes were obtained (Fig. S11–S12). The sequence analyses of PCR products of each PCR–RFLP amplicon identified gene cassettes with amplicon sizes: 1687 bp, 1597 bp, and 792 bp. The integrons carried aminoglycoside adenyl transferase (*aadA1* and *aadA5*) conferring resistance to streptomycin-spectinomycin and dihydrofolate reductase (*dfrA7*, *dfrA17*, and *dfrA1*) providing resistance against trimethoprim (Fig. 3). The 75% (27/33) of the isolates contained 1687 bp integron gene cassette composed of (*dfrA17* + *aadA5*) while 11% (4/33) isolates carried gene cassette of size 792 bp with *dfrA7* and 5% (2/33) contained gene cassette with composition (*dfrA1* + *aadA1*) of size 1597 bp. The sequencing results are submitted to NCBI GenBank under accession numbers OM812939, OM812947, and OM812948. In 97% (35/36) of *intI1* positive isolates, the 3` conserved segment was found to be associated with *sul1* and *qacΔE1* conferring

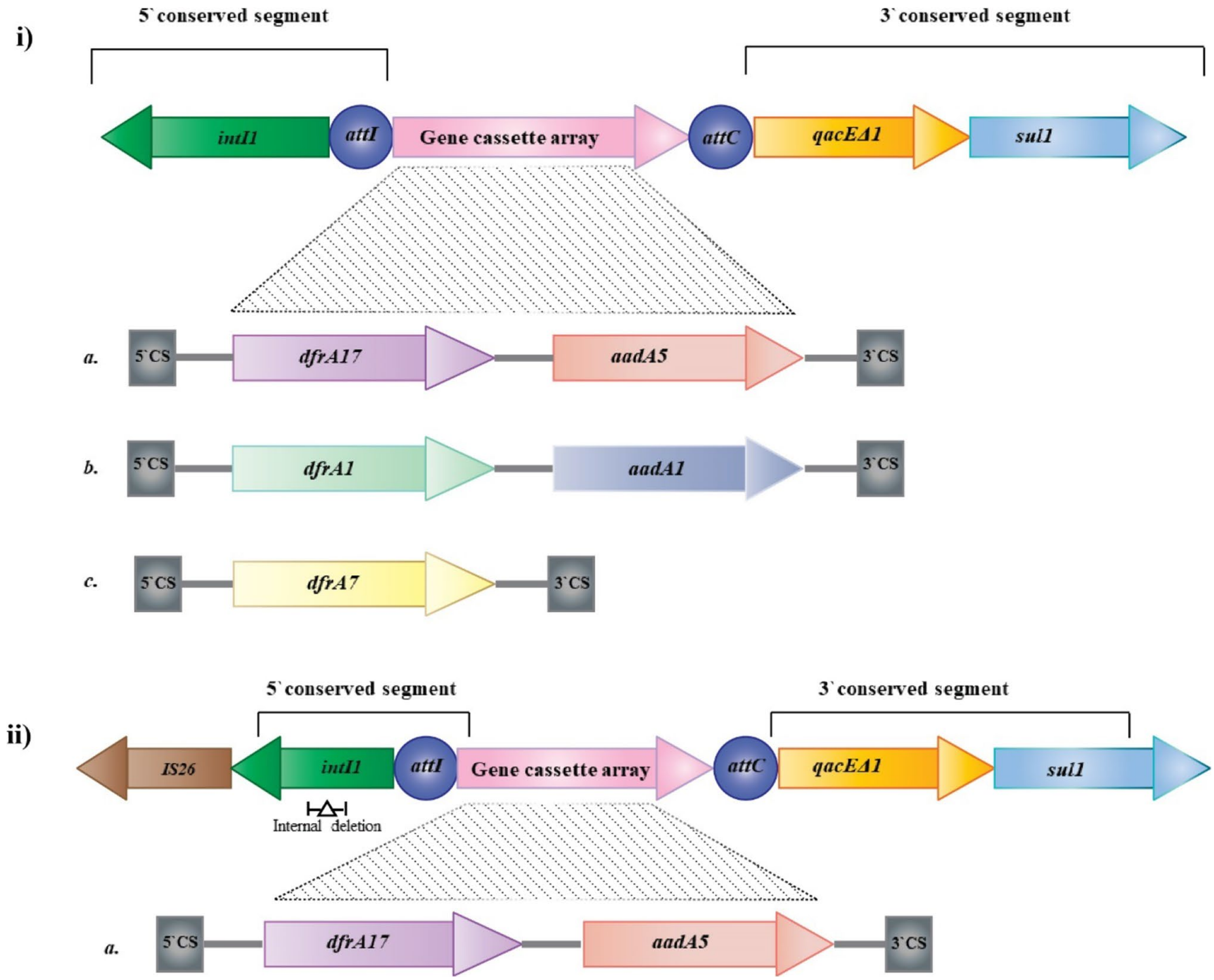

Fig. 3 Schematic representation of the various class 1 integron gene cassettes identified in *E. coli* isolates from bovine mastitis. The direction of arrows indicates the direction of transcription for each gene. (i) Representation of three different types of gene cassettes identified. (a), (b) display 1600 bp gene cassette and (c) display 800 bp gene cassette. (ii) Representation of class 1 integrase linked to IS26 with an internal deletion of 265 bp associated with gene cassette *dfrA17*+*aadA5*

resistance against sulfonamides and quaternary ammonium compounds.

Novel variants of aadA5 and dfrA17

The sequence analysis of 12 randomly selected isolates carrying gene cassette with *dfrA17*+*aadA5* revealed 4 isolates with a new variant of *aadA5* named *aadA5.1*, *aadA5.2*, *aadA5.3*, and *aadA5.4* (Figs. S13–S14). There were 2 isolates with a new variant of *dfrA17* named *dfrA17.1*, and *dfrA17.2* (Figs. S15–S16). The sequencing results are submitted to NCBI GenBank under accession numbers OM812939, OM812940, OM812941, OM812942, OM812943, OM812944, OM812945, OM812946, OM812949, OM812950, OM812951, OM812952, OM812953, and OM812954. Isolates with accession number OM812939, OM812940, OM812941, OM812942, OM812943, and OM812946 carried novel alleles.

Among the 4 isolates carrying a new allele in the *aadA5* gene, *aadA5.1*, *aadA5.2*, and *aadA5.3* carried a new allele in their main functional domain nucleotidyltransferase domain (Fig. 4) and *aadA5.4* had it in the domain of the unknown function (DUF4111). A homology check of the sequence revealed substitution K43R in *aadA5.1*, (I31L + L33A) in *aadA5.2*, (I31M + D45V) in *aadA5.3* and F122S in *aadA5.4*. One of the isolates harbored a variant of *aadA5* with substitution E235G (*aadA5.5*), which has been reported earlier in NCBI under accession number HBF3136034. Substitutions observed in *dfrA17.1* is R45G followed by I80V in *dfrA17.2* (Fig. 5). To further compare the activity of *dfrA17*

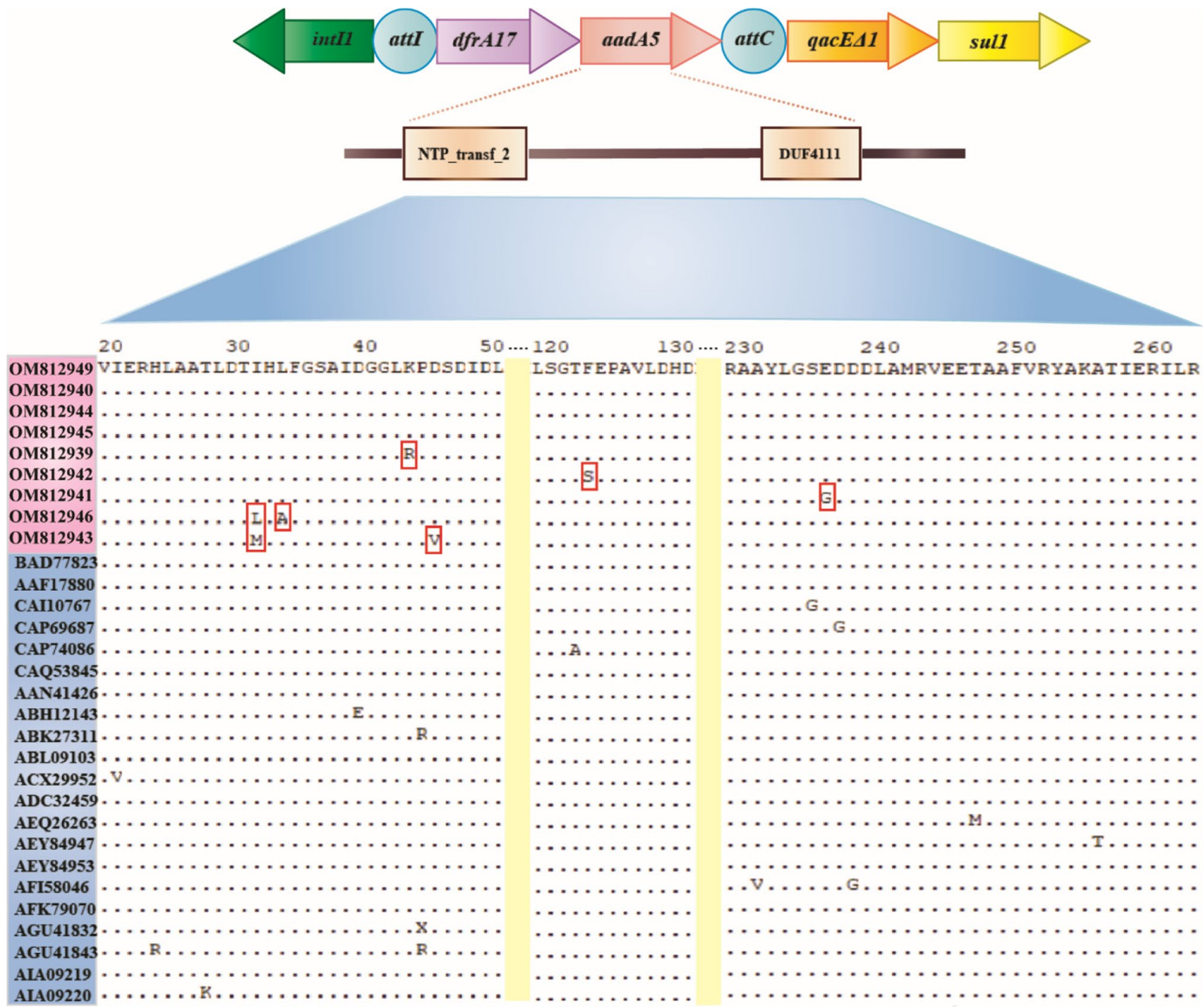

Fig. 4 Schematic representation of aminoglycoside adenyl transferase (*aadA5*) with multiple sequence alignment. The sequences highlighted in blue indicate earlier sequences submitted and in pink are sequences obtained in this study. The novel alleles of *aadA5* discovered in this study are enclosed in red boxes

new variants and the native gene MIC were checked. The isolates carrying *dfrA17.1*, *dfrA17.2,* and native *dfrA17* showed the same MIC of 240 µg/ml (Fig. S17).

Prediction of deleterious and tolerant substitutions by SIFT and PROVEAN

Among the seven mutations found in the *aadA5* gene, two were predicted to be deleterious with substitution (L33A), and (F122S) according to scores of both the SIFT and PROVEAN software. In the *dfrA17* gene, the substitution (R45G) was predicted to be deleterious (Table 2). The deleterious mutations are the genetic alternations that affect the protein function. There were five *aadA5* and one *dfrA17* mutation in the tolerated category. The tolerant mutations show a lower impact on protein function.

Molecular docking of native aadA5 and its variants

The docking structure of A chain of native *aadA5* receptors binds with ligand streptomycin with the binding affinity of − 7.1 kcal/mol. However, the *aadA5* variants showed a decrease in binding affinity with a value of − 7.0 kcal/mol, indicating that mutations do affect the binding between streptomycin and *aadA5*.

The docking results demonstrated that streptomycin interacts with native *aadA5* and variants *aadA5.2* and *aadA5.3* through eight hydrogen bonds (Table 3). The common amino acid residues involved in hydrogen

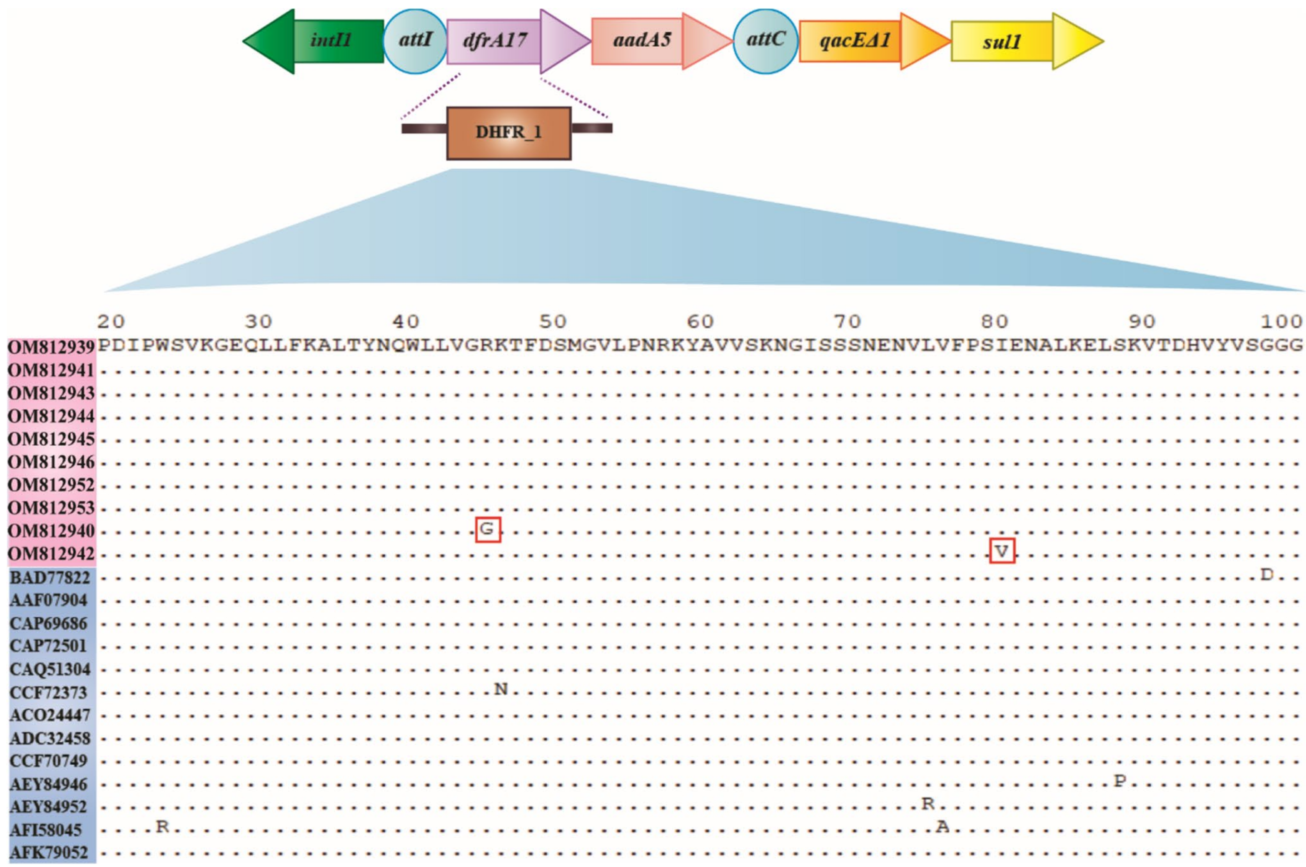

Fig. 5 Schematic representation of dihydrofolate reductase (*dfrA17*) with multiple sequence alignment. The sequences highlighted in blue indicate earlier sequences submitted and in pink are sequences obtained in this study. The novel alleles of *dfrA17* discovered in this study are enclosed in red boxes

Table 2 SIFT and PROVEAN prediction results for novel variations identified in *aadA5* and *dfrA17*

S.No	Gene	Amino acid variation	SIFT		PROVEAN	
			Score	Prediction	Score	Prediction
1	*aadA5*	K43R	0.46	Tolerated	− 1.580	Neutral
2	*aadA5*	I31L	0.23	Tolerated	− 1.830	Neutral
3	*aadA5*	I31M	0.06	Tolerated	− 2.365	Neutral
4	*aadA5*	L33A	0.02	Affect protein function	− 4.632	Deleterious
5	*aadA5*	D45V	0.06	Tolerated	− 1.128	Neutral
6	*aadA5*	F122S	0.04	Affect protein function	− 5.031	Deleterious
7	*aadA5*	E235G	0.41	Tolerated	− 0.756	Neutral
8	*dfrA17*	R45G	0.00	Affect protein function	− 6.972	Deleterious
9	*dfrA17*	I80V	1.00	Tolerated	− 0.753	Neutral

bond interactions involved (GLU87, ARG192, GLN172, ASP130, ARG105, ASP128, LYS137) have been represented in Fig. 6. In variants *aadA5.1*, *aadA5.4*, and *aadA5.5*, the number of hydrogen bonds was reduced, but still occurred in (GLU87, LYS205) (Stern et al. 2018) and can be observed in the 2D interaction diagram (Fig. S18).

Discussion

This study investigated the prevalence of third generation cephalosporins and fluoroquinolones resistant *E. coli* in bovine mastitis and their multi drug resistance pattern

Table 3 Result of molecular docking between native and novel variants of *aadA5* and the ligand streptomycin

Docking structure	Affinity (kcal/mol)	Interacting amino acids of *aadA5* protein
Native *aadA5* with streptomycin	−7.1	GLU87, ARG192, GLN172, GLN108, ASP130, ARG105, ASP128, LYS137
aadA5.1 (K43R) with streptomycin	−7.0	GLU87, LYS205, GLN172, GLN108
aadA5.2 (I31L+L33A) with streptomycin	−7.0	GLU87, ARG192, GLN172, ASP130, ARG105, ASP128, LYS137, LEU127
aadA5.3 (I31M+D45V) with streptomycin	−7.0	GLU87, ARG192, GLN172, GLN108, ASP130, ARG105, ASP128, LYS137
aadA5.4 (F122S) with streptomycin	−7.0	ASP49, LYS205, GLN172, GLN108, ASP130, ASP128
aadA5.5 (E235G) with streptomycin	−7.0	GLU87, LYS205, GLN172, GLN108, ASP130

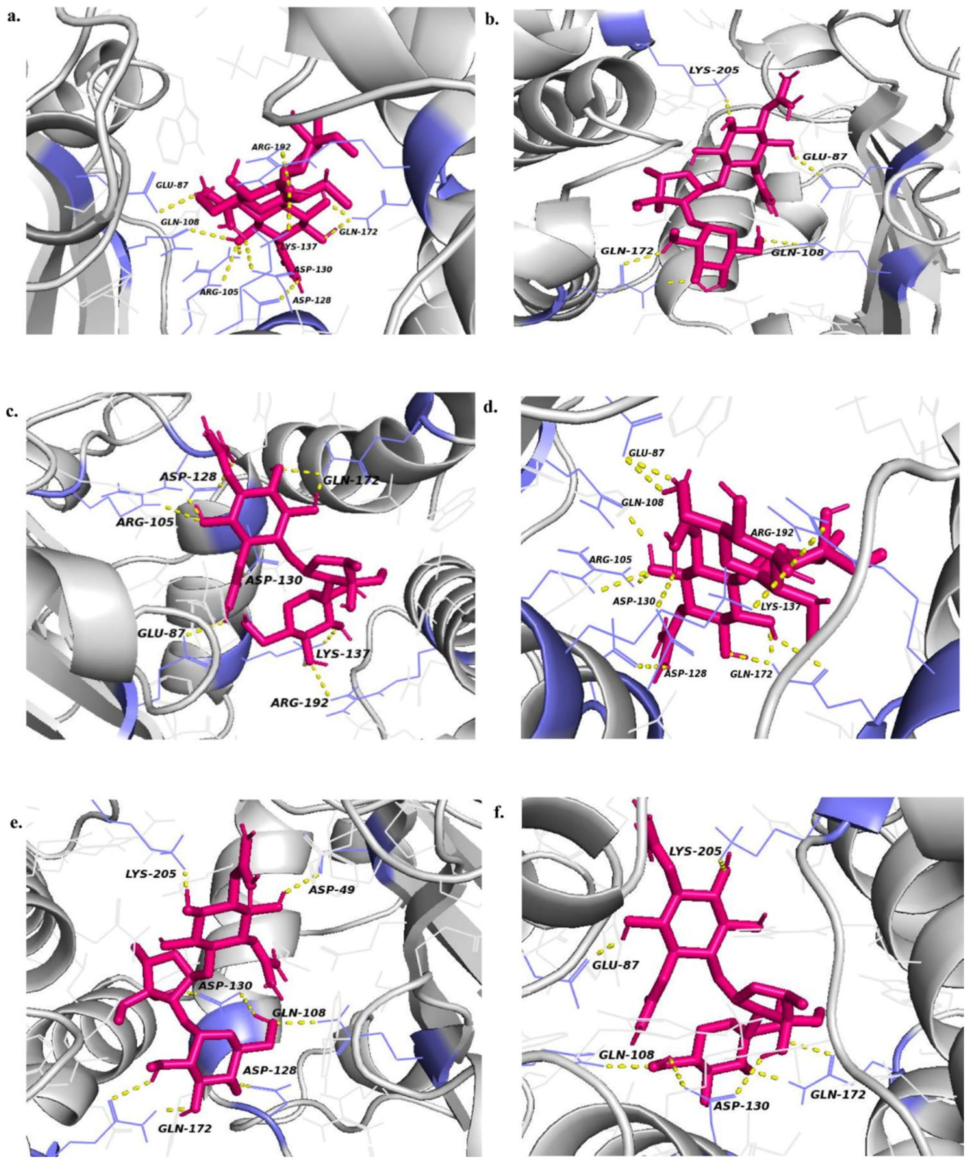

Fig. 6 Molecular docking results. 3D representation of *aadA5* and its novel variants with streptomycin. **a** Native *aadA5*, **b** *aadA5.1*, **c** *aadA5.2*, **d** *aadA5.3*, **e** *aadA5.4*, **f** *aadA5.5*. Yellow dotted lines represent hydrogen bond interactions

conferred by mobile genetic elements.

In this study among the 300 clinical mastitis milk samples, 110 *E. coli* isolates were found phenotypically resistant to fluoroquinolones and cephalosporins. We detected 100% isolates phenotypically resistant to cefotaxime and 98% of them carrying ESBL genes. The major ESBL gene detected in our study was $bla_{\text{CTX-M}}$, with $bla_{\text{CTX-M-1}}$ group being the most prevalent gene group. The occurrence of multiple ESBL genes in *E. coli* strains has been earlier observed in humans and livestock (Leverstein-van Hall et al. 2011). This study also detected 74% (82/110) of the isolates carrying multiple ESBL genes, which display an array of MIC patterns according to different gene combinations (Fig. S9). This predominance of ESBL genes among bovine mastitis suggests higher spread and usage of third generation cephalosporins in veterinary. Although only 16 isolates were found to carry ESBL genes alone. Genotypic characterization detected 83% (92/110) of *E. coli* isolates resistant to both β-lactam and fluoroquinolones which suggests high consumption of these antibiotics in livestock. The presence of ESBL + QRDR, ESBL + PMQR + QRDR, and ESBL + PMQR occurred in 39%, 37%, and 7% of isolates, respectively (Fig. 2). This study indicates QRDRs as a major fluoroquinolone resistant determining factor as compared to PMQR which agrees with previous studies (Qian et al. 2020). According to previous studies, *gyrA*(S83L/D87N) or *parC*(S80I) has been reported as the most prevalent QRDR mutations in mastitis (Yang et al. 2018) which is contradictory to our reports which suggest *gyrB*(S492N) as the main contributor to bovine mastitis. This study reports the co-existence of PMQR genes with ESBL in 44% (49/110) isolates. Among which 93% (41/44) isolates carried $bla_{\text{CTX-M-1}}$ and bla_{TEM} in combination with PMQR genes. This implies co-occurrence of ESBL genes $bla_{\text{CTX-M-1}}$ and bla_{TEM} with PMQR in the same plasmid plays a crucial role in their co-transmission as reported earlier (Tao et al. 2020). The existence of *E. coli* strains co-resistant to ESBL and fluoroquinolones have been detected in all ecosystems including wastewater/river (Conte et al. 2017), human/clinical (Azargun et al. 2018), animals (Eisenberger et al. 2018), and soil (Kimera et al. 2021). This can be related to the cross-movement of isolates or the ARGs between different ecosystems facilitated by the spread of mobile genetic elements. In this study, IS*26* emerged as the most prevalent MGE with 70% (76/110) of *E. coli* isolates harboring it and found physically linked to *intI1* in 17% of isolates. This imparts support to growing documentation of IS*26* as a plausible means of ARGs dissemination in pathogenic *E. coli* (Xie et al. 2018; Oliva et al. 2018). The sequencing of ~ 1200 bp IS*26* and *intI1* linked amplicons displayed IS*26* is present upstream to *intI1* and in the same direction as it (Fig. S19). In the isolates IS*26* is linked to *intI1* an internal deletion of 265 bp was observed in the *intI1* gene (Fig. S20) earlier reported in *Escherichia coli* strain ST131 isolates from human urine, blood, and feces samples and submitted with accession numbers MK295819, CP041573, AP018456, LC520288, CP054458 from countries the USA, China, and Japan in NCBI. This study reports IS*26* linked to *intI1* with an internal deletion in 5`CS of integrase gene in bovine mastitis for the first time. Earlier a deletion in 3`CS of class 1 integron has been reported in *E. coli* isolates from human and animal sources (Dawes et al. 2010). This indicates local and global transmission of plasmids, MGEs, and ARGs between humans and animals as reported earlier (Kim en Cha 2021). This linkage may result in the co-selection of IS*26*, *intI1*, and the ARGs associated with class 1 integron gene cassettes.

This study reports class 1 integrons carrying three different types of gene cassettes. These gene cassettes harbor ARGs against streptomycin/spectinomycin and trimethoprim. The presence of integrons resulted in the emergence of 35 *E. coli* isolates as multidrug resistant (MDR) carrying antibiotic resistant genes against three or more antibiotic classes. The physical linkage between ARGs in integron gene cassettes, *qacΔE1*, and *sul1* genes implies a significant possibility for the emergence of MDR bacteria in a fixed population and their transmission within organisms. The presence of *qacΔE1* and *sul1* genes independent of class1 integrase gene was observed in 7 isolates. In total, this study reports 40% of *E. coli* isolates from mastitis milk samples as MDR.

This study reports the prevalence of (*dfrA17* + *aadA5*) gene cassette among class 1 integron positive isolates. This indicates the high usage of trimethoprim and streptomycin/spectinomycin antibiotics in livestock leading to the low fitness cost of this gene cassette. Similar findings of the abundance of aminoglycoside and trimethoprim resistant gene cassettes have been reported in both humans and animals (Ma et al. 2017). Also, 70% of the isolates carrying (*dfrA17* + *aadA5*) gene cassette had IS*26* linked to class 1 integron facilitating their spread across the organisms. A detailed study of this gene cassette led to the determination of novel variants of the *dfrA17* and *aadA5* genes. Molecular docking studies of *aadA5* novel variants displayed a decrease in binding affinity as compared to the native protein. Which signifies the strong interaction of the native protein with streptomycin. The results unveiled the interaction of critical amino acids Glu87, Arg192, Asp130, and Lys205 with streptomycin in native *aadA5* and all novel variants except in *aadA5*.4 with mutation (F122S). The *aadA5*.4 variant did not show interaction with Glu87 the main residue involved in the adenylation of streptomycin to inhibit its activity(Stern et al. 2018). Also, the potential effect of mutation studied by SIFT and PROVEAN showed F122S and L33A as deleterious SNPs affecting protein function. This study identifies novel variants *aadA5*.1, *aadA5*.2, and *aadA5*.3 as neutral mutations exhibiting the same antibiotic activity as native *aadA5*. While Insilco studies display *aadA5*.4 with reduced activity as compared to the native protein.

One major finding of this study is *aadA5*.5 with substitution E235G reported for the first time in the Indian subcontinent. The sequence for the same is submitted in NCBI under accession number OM812941. This variant has been earlier reported only in *Shigella sonnei* isolated from the human clinical samples in Belgium. This study displays the transferability of ARGs and resistant plasmids conquering inter-kingdom barriers between humans and animals (Fig. S21). This breaching of taxonomic barriers can be attributed to the presence of resistant plasmids IncFIA, IncFIB, IncFII, IncI1, and IS elements like IS*26* which are a major source of horizontal gene transfer (Vrancianu et al. 2020). Another important factor is the presence of class 1 integrons which are spread over a large number of bacterial species and can translate antibiotic resistance across a wide range of hosts with low negative selection pressure due to high usage of antimicrobials by humans (Gillings 2018). This study reports the transfer of ARGs associated with MGEs from humans to the bovine environment which can ultimately serve as a potential reservoir and risk for human health.

This study shows the prevalence of class 1 integrons and IS elements in MDR *E. coli* from bovine mastitis. A high incidence of fluoroquinolone resistance in ESBL-producing *E. coli* was reported in the study. *E. coli* isolates were investigated for the presence of class 1 integron gene cassettes and were identified with four novel alleles of aminoglycoside adenyl transferase (*aadA5*) and two of dihydrofolate reductase (*dfrA17*). The environmental reservoir of ARGs plays an important role in transmission associated with MGE between animals, humans, and the environment. Also, the higher use of antibiotics may encourage the selective enrichment of specific antibiotic resistant bacteria across ecosystems. Proper waste disposal is critical to breaking the transmission of mobile genetic elements from human to animal and vice versa.

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Data availability All data generated or analyzed during this study are included in this published article (and its supplementary information files).

Declarations

Ethics approval This article does not contain any studies with human participants or animals performed by any of the authors.

Conflict of interest The authors declare no competing interests.

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